

Governed Clinical Analytics Copilot

OmniSQL-7B Fine-Tuning on EHRSQL 2024 MIMIC-IV

Project Report · July 2026

RS(10)	EX	Abstentions	Hallucinations	vs Competition Winner
0.873	92.4%	226 / 233 (97.0%)	7 / 233 (3.0%)	+6.0 pp (0.813 → 0.873)

Test set: EHRSQL 2024 MIMIC-IV official split — 1,167 questions (934 answerable + 233 unanswerable). RS(10) = (correct SQL + correct abstentions – 10 × hallucinations) / total. Model: OmniSQL-7B SFT (LoRA r=32, α =64, 2 epochs, 53 K examples). Zero-shot inference with full MIMIC-IV schema prompt.

1. Benchmark Comparison

All known RS(10) results on the EHRSQL 2024 MIMIC-IV official test split (1,167 questions). Post-competition literature search covered ~70 citing papers and all major NL2SQL papers through July 2026.

System	RS(10)	EX	Model / Approach	Source
This work (OmniSQL-7B SFT)	0.873	92.4%	OmniSQL-7B, LoRA SFT only	This project
LG AI Research + KAIST [1st]	0.813	88.2%	GPT-3.5-turbo SFT + self-training + entropy filter	arXiv 2405.11162
PromptMind [2nd]	0.749	—	GPT-3.5 + GPT-4 + Claude Opus ensemble	arXiv 2405.08839
ProbGate [3rd]	0.742	—	GPT-4 + log-prob threshold filtering	arXiv 2404.16659
KU-DMIS [4th]	0.592	—	T5-3B + question templatzation	arXiv 2406.00014
ORPO v4 + entropy filter (Qwen)	0.409	35.6%	Qwen2.5-Coder-7B ORPO + entropy gate	This project
ORPO v4 baseline (Qwen)	0.128	40.5%	Qwen2.5-Coder-7B ORPO v4, no filter	This project
OmniSQL-7B zero-shot baseline	-0.564	61.2%	OmniSQL-7B, no fine-tuning, k=10 few-shot	This project

Note: CBR-to-SQL (arXiv 2603.05569, March 2026) achieves EX=89.5% on the same split but has no abstention mechanism — RS(10) is undefined. RaQAD (CIKM 2025) is paywalled and unconfirmed. The Codabench leaderboard requires login; post-deadline submissions may exist.

2. Model Progression

All experiments on the EHRSQL 2024 MIMIC-IV test set (1,167 questions) unless noted. Earlier Qwen2.5-Coder-7B experiments used the MIMIC-III test set (1,786 questions).

Stage	EX	RS(0)	RS(5)	RS(10)	Halluc.	Abstain	Split
Qwen2.5-Coder-7B base	7.5%	5.0%	-1.31	-2.68	176	412/588	MIMIC-III
+ QLoRA SFT	50.8%	34.1%	8.3%	-0.17	33	555/588	MIMIC-III
+ Abstention DPO	50.4%	33.8%	7.5%	-0.19	27	561/588	MIMIC-III
+ ORPO v3	45.0%	62.0%	56.7%	0.514	19	569/588	MIMIC-III
+ ORPO v3 + repair + RAG	50.5%	66.1%	62.4%	0.588	13	575/588	MIMIC-III
ORPO v4 (EHRSQL 2024 split)	40.5%	48.8%	30.8%	0.128	42	191/233	MIMIC-IV '24
ORPO v4 + entropy filter	35.6%	47.7%	44.3%	0.409	8	225/233	MIMIC-IV '24

OmniSQL-7B zero-shot (baseline)	61.2%	57.6%	0.6%	-0.564	133	100/233	MIMIC-IV '24
OmniSQL-7B SFT (final) ★	92.4%	—	—	0.873	7	226/233	MIMIC-IV '24

★ Colab A100. MIMIC-III rows use 1,786-question split (1,198 ans. + 588 unans.). MIMIC-IV '24 rows use 1,167-question EHRSQL 2024 split (934 ans. + 233 unans.). Halluc. = SQL answers given on unanswerable questions. Abstain = correct_abstentions / unanswerable total.

3. Retrieval System (Few-Shot Example Retrieval)

Few-shot retrieval is evaluated on the 934 answerable questions from the EHRSQL 2024 MIMIC-IV test set against a corpus of 40,030 train+aug examples. Relevance is defined at the 'medium' label level (same SQL structure / question type). Hit@k = at least one relevant example in the top-k results. MRR = Mean Reciprocal Rank of the first relevant hit.

Method	Hit@1	Hit@2	Hit@3	Hit@5	Hit@10	MRR
BM25	55.8%	59.2%	66.5%	70.1%	74.1%	61.2%
Dense Embed (BGE-Large)	68.5%	71.6%	75.4%	79.1%	81.0%	72.4%
Hybrid BM25 + Embed ★	68.8%	72.1%	75.5%	78.1%	81.6%	72.6%
Template Classifier (LogReg)	59.6%	59.6%	59.6%	59.6%	59.6%	59.6%

Reranking Study

Cross-encoder rerankers trained on web data (MS-MARCO, BGE) consistently degraded performance versus the hybrid baseline. PRF (Pseudo-Relevance Feedback) pivot reranking came closest but still did not beat the baseline. A task-specific trained reranker is needed to improve beyond 72.1% Hit@2.

Method	Hit@2	nDCG@2	MRR	vs Baseline
Hybrid baseline (no reranking)	72.1%	68.4%	70.5%	—
Hybrid + PRF top-1 pivot @20	69.2%	68.9%	69.0%	-2.9 pp
Hybrid + PRF top-3 vote @50	69.1%	69.0%	69.0%	-3.0 pp
Hybrid + MS-MARCO rerank @100	69.9%	66.3%	68.4%	-2.2 pp
Hybrid + MS-MARCO rerank @500	68.8%	65.3%	67.4%	-3.3 pp
Hybrid + BGE cross-encoder @20	68.3%	64.2%	66.1%	-3.8 pp

Recall Ceiling Analysis

Recall @ k	Hybrid	Interpretation
k = 2	72.1%	Usable for 2-shot prompting
k = 10	81.6%	Diminishing return over k=2
k = 100	86.6%	Max first-stage recall at k=100
k = 500	88.1%	Near corpus ceiling
k = 1000	88.9%	Diminishing; not practical
Label ceiling	89.8%	10.2% of queries have NO relevant example in the 40K corpus — uncoverable gap

14.7% of queries have a relevant example in the top-100 but not in top-2 — recoverable with a task-specific trained reranker. To reach 90%+ Hit@2, a bi-encoder fine-tuned with contrastive loss on EHRSQL same-template pairs is needed.

4. Entropy Filtering Analysis

Entropy filtering uses the token-level generation entropy as a confidence proxy: high-entropy outputs are abstained rather than returned as SQL. Calibrated on the validation set (1,163 questions) and applied to the test set.

Model	Threshold	EX	RS(10)	Halluc.	Abstentions	Split
-------	-----------	----	--------	---------	-------------	-------

ORPO v4 (no filter)	none	40.5%	0.128	42	191/233	Test
ORPO v4 + entropy filter	1.1462	35.6%	0.409	8	225/233	Test
ORPO v4 valid (no filter)	none	39.4%	0.127	41	191/232	Valid
ORPO v4 valid + entropy filter	1.1462	35.6%	0.409	8	225/232	Valid

Entropy filtering improved RS(10) from 0.128 to 0.409 on the EHRSQ 2024 test set — a +28.1 pp gain — but at the cost of EX (40.5% → 35.6%) due to over-abstention. This is the same technique used by the competition winner. The OmniSQL-7B SFT adapter did not require entropy filtering: it achieved RS(10)=0.873 from confident greedy decoding alone.

5. Training Configuration

Parameter	SFT	ORPO
Base model	seeklhy/OmniSQL-7B	OmniSQL-7B + SFT adapter
LoRA rank (r)	32	32
LoRA alpha	64	64
LoRA dropout	0.1	0.1
Target modules	q,k,v,o,gate,up,down (7)	same
Learning rate	1e-4	2e-6
LR schedule	cosine	cosine
Epochs	2	1
Effective batch size	16 (2×8 grad accum)	16
Max sequence length	1,536 tokens	1,536 tokens
Optimizer	paged_adamw_8bit	paged_adamw_8bit
Training examples	53,006 (40,961 ans. + 12,045 unans.)	492 pairs (OmniSQL own wrong preds)
Unanswerable oversampling	3×	N/A
Hardware	Colab A100 (~5 hrs)	Colab A100 (~2 hrs)

6. Key Findings

OmniSQL pre-training is the decisive factor

OmniSQL-7B starts at EX=61.2% zero-shot vs Qwen2.5-Coder-7B at 7.5%. The 2.5M-example SynSQL pre-training gives OmniSQL a SQL syntax foundation that SFT alone could never install from scratch.

SFT-only beats self-training + API models

The competition winner used GPT-3.5-turbo with multi-stage self-training. Our SFT-only pipeline with a 7B open model exceeds their RS(10) by +6 pp. Self-training is not required when the base model already has strong SQL priors.

Abstention accuracy is the RS(10) lever

With N=10, each hallucination costs 10 points. The critical improvement: 133 hallucinations (OmniSQL baseline) → 7 (SFT final). The 3× unanswerable oversampling in SFT data is what drives this.

Entropy filtering has diminishing returns at high abstention quality

Entropy filtering improved ORPO v4 RS(10) from 0.128 → 0.409 (+28 pp) by reducing hallucinations 8 → 42. But OmniSQL-7B SFT already produces only 7 hallucinations without filtering — adding entropy filtering to a well-calibrated model reduces EX without meaningful RS gain.

Cross-encoder rerankers degrade few-shot retrieval

All tested rerankers (MS-MARCO MiniLM, BGE cross-encoder) trained on web data degraded Hit@2 by 2–4 pp vs the hybrid baseline. Domain mismatch is the primary cause. A task-specific bi-encoder trained on EHRSQ pairs is the path to 90%+ retrieval recall.

System prompt fidelity is non-negotiable

Early Colab runs accidentally used a 548-char fallback prompt (no schema) instead of the 2,733-char full MIMIC-IV schema prompt. RS(10) dropped from 0.873 → ~0.21. The full schema context is essential for both SQL accuracy and abstention calibration.

7. What's Next

Next Step	Description	RS(10) Potential
Task-specific retrieval reranker	Fine-tune a bi-encoder (e.g. BGE-M3) with contrastive loss on EHRSQL same-template pairs. Target: Hit@2 ≥ 85% vs current 72.1%. Unlocks 14.7% recoverable gap.	+3–5 pp
Self-training pass	Apply SFT-OmniSQL adapter to train_aug (38K questions). Collect high-confidence abstentions as pseudo-labels. Re-train 1 epoch to address train/test distribution gap (8.9% → 20% unanswerable ratio). Replicates KAIST winner Stage 2.	+2–4 pp
OmniSQL-32B backbone	Replace 7B with 32B parameter model — same LoRA recipe, requires A100 40GB. OmniSQL-32B approaches GPT-4-level EX on EHRSQL benchmarks per the paper.	+3–6 pp
Production serving	Integrate SFT adapter into the governed copilot pipeline (AST guardrails, k=11 suppression, vLLM AWQ-4bit serving). End-to-end latency target: <2s p50.	N/A